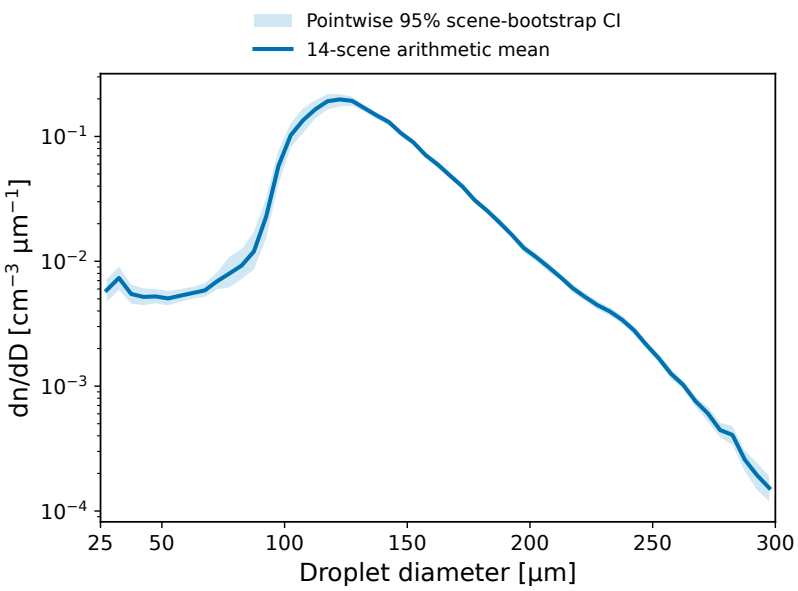
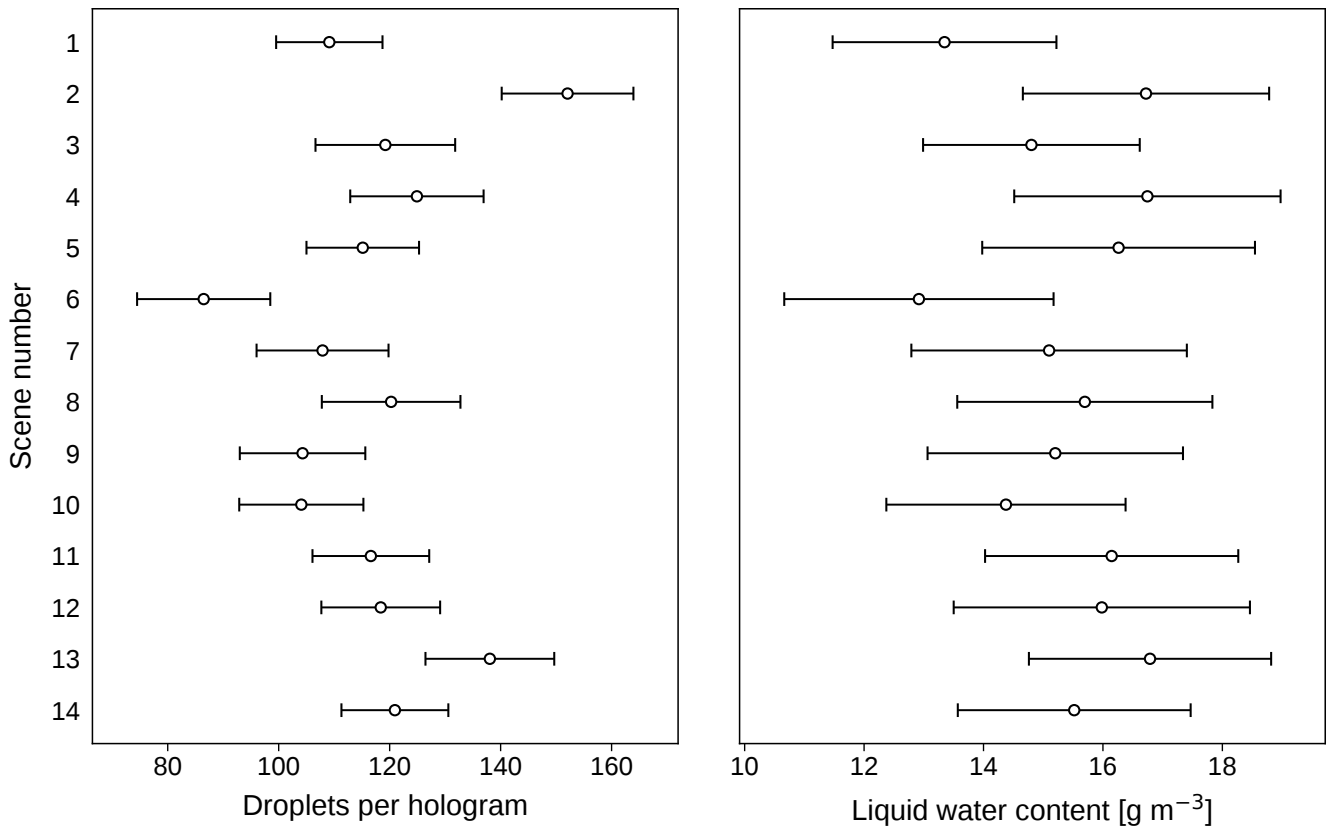
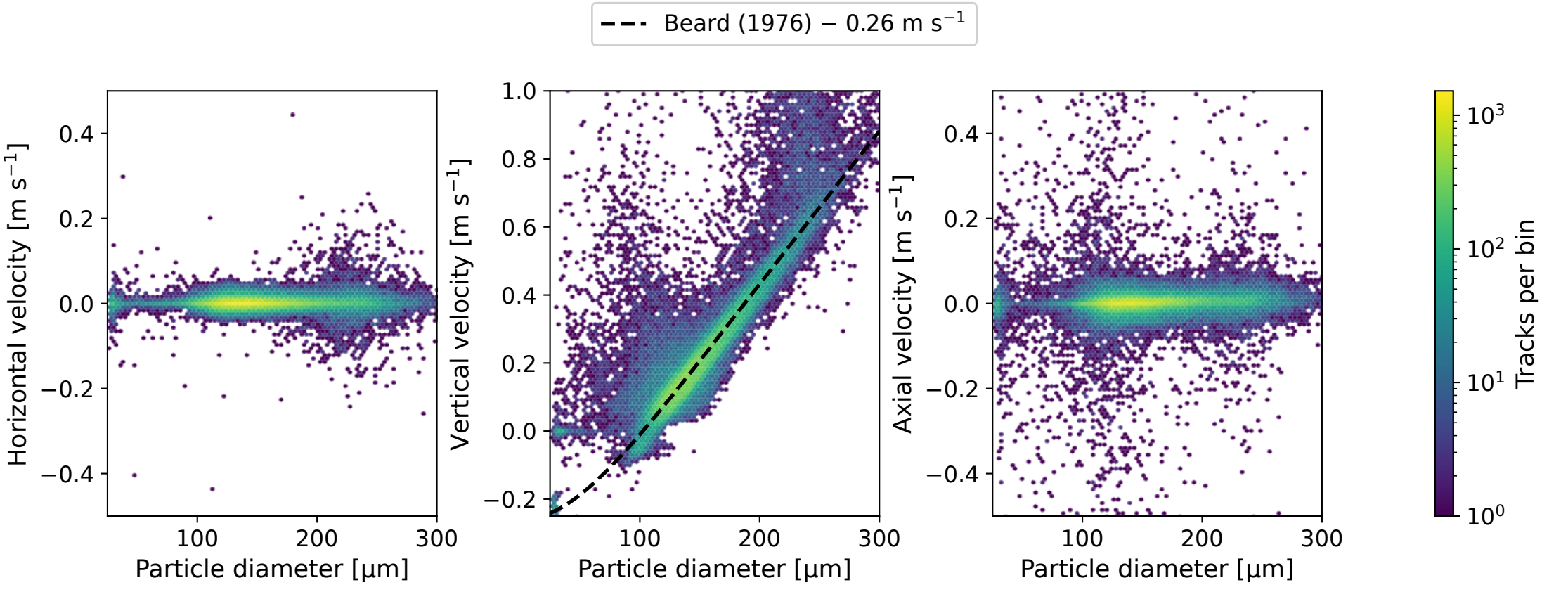


(a) Number-density distribution



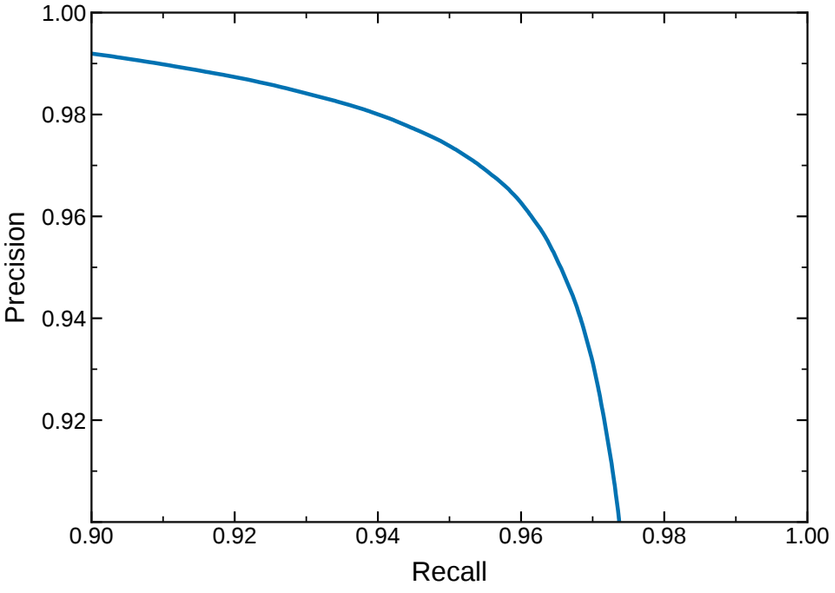
(b) Droplet count and liquid water content



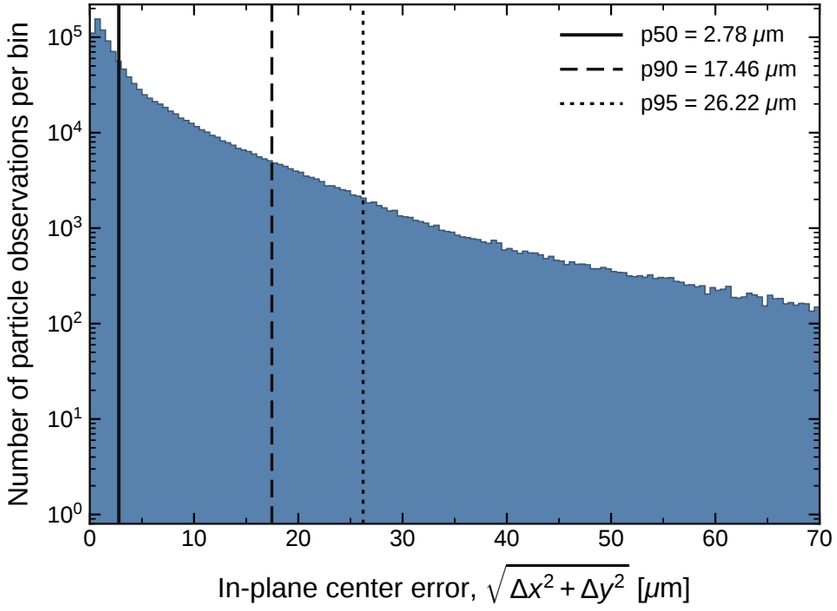
(a) Horizontal velocity

(b) Vertical velocity (downward positive)

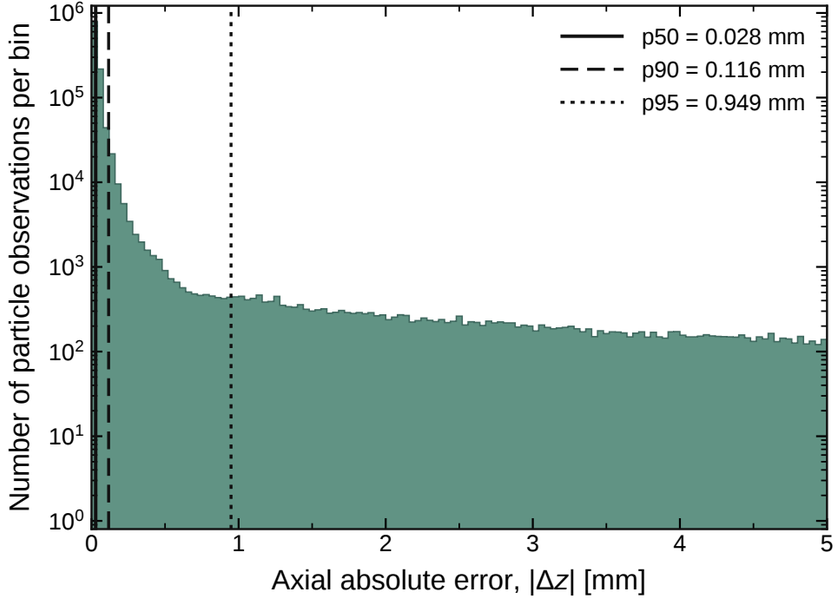
(c) Axial velocity


Precision
Recall
1.00
0.98
0.96
0.94
0.92
0.90
0.92
0.94
0.96
0.98
1.00

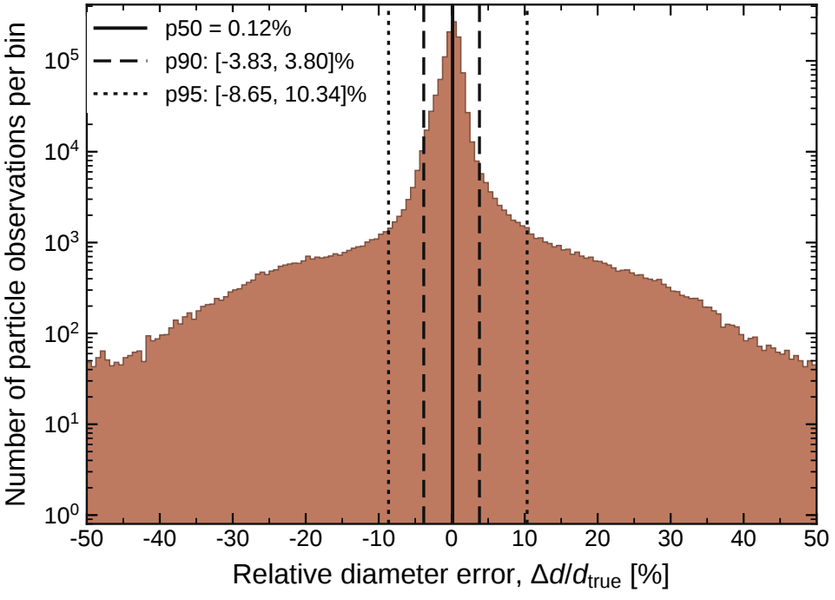
(a)


p50 = 2.78 μm
p90 = 17.46 μm
p95 = 26.22 μm
Number of particle observations per bin
In-plane center error, $\sqrt{\Delta x^2 + \Delta y^2}$ [μm]

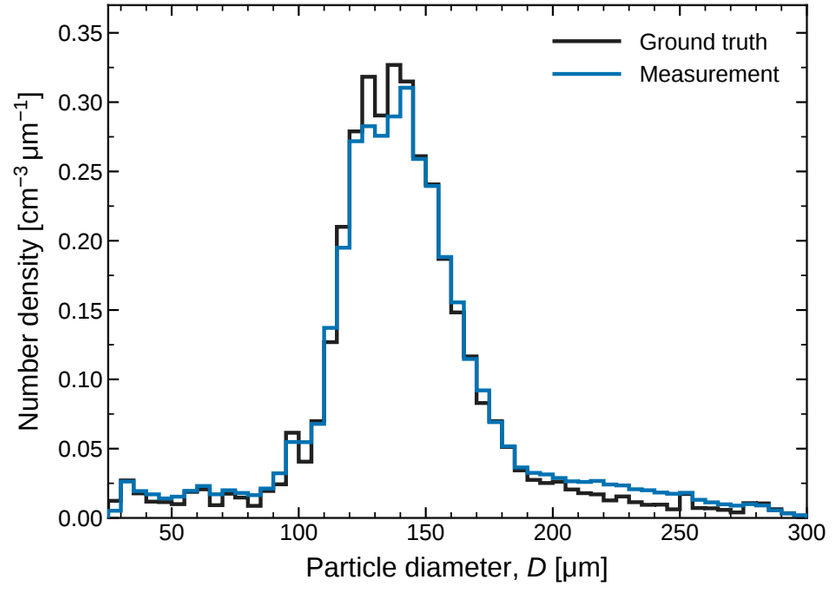
(b)


p50 = 0.028 mm
p90 = 0.116 mm
p95 = 0.949 mm
Number of particle observations per bin
Axial absolute error, |Δz| [mm]

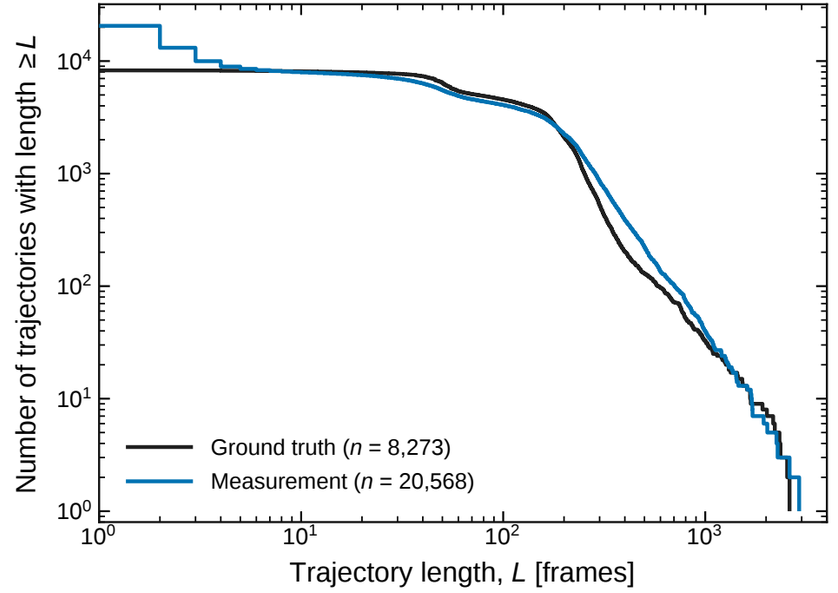
(c)


p50 = 0.12%
p90: [-3.83, 3.80]%
p95: [-8.65, 10.34]%
Number of particle observations per bin
Relative diameter error, $\Delta d/d_{\text{true}}$ [%]

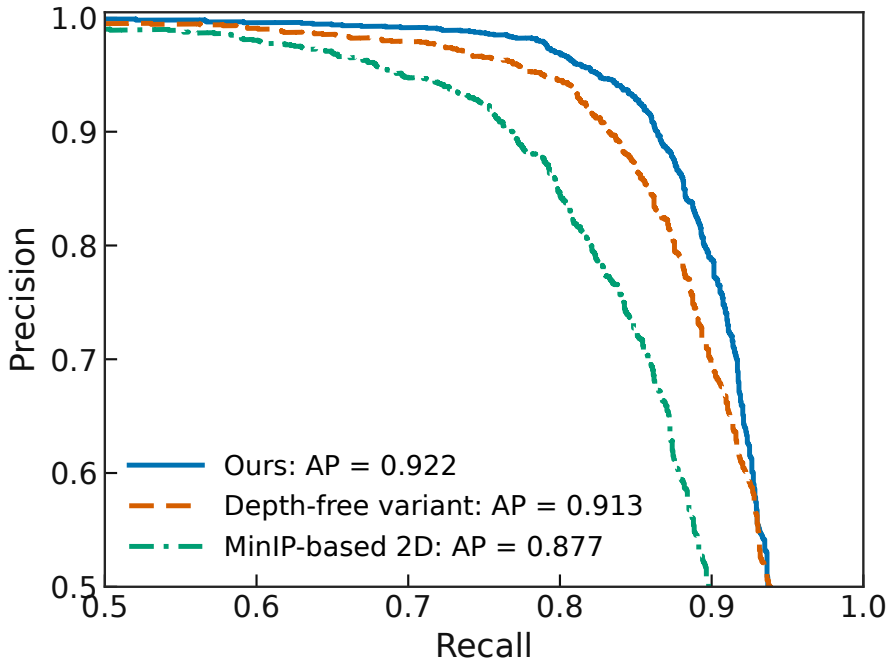
(d)


Ground truth
Measurement
Number density [cm^{-3} μm^{-1}]
Particle diameter, D [μm]

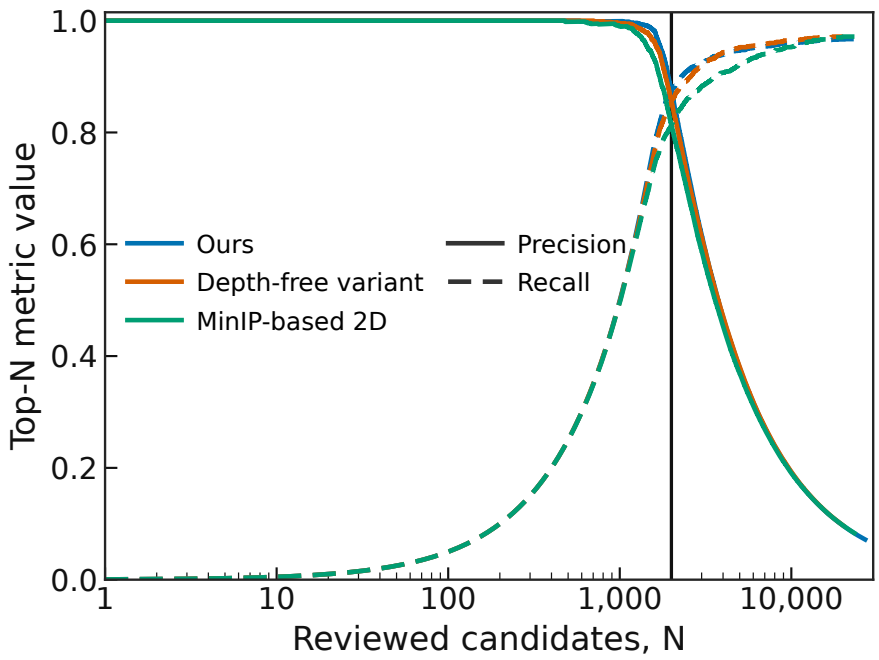
(e)


Ground truth (n = 8,273)
Measurement (n = 20,568)
Number of trajectories with length ≥ L
Trajectory length, L [frames]

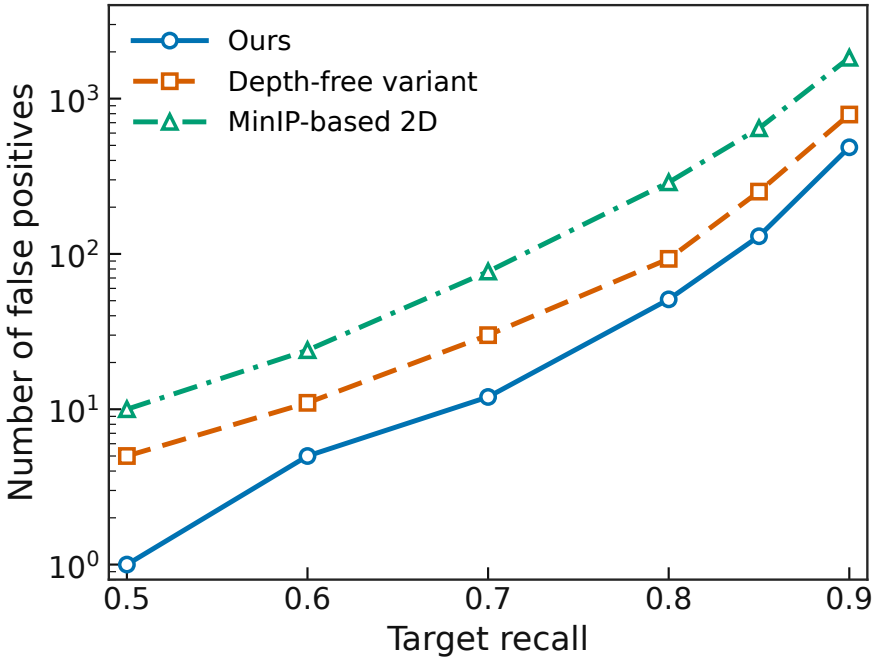
(f)


Precision
Recall
Ours: AP = 0.922
Depth-free variant: AP = 0.913
MinIP-based 2D: AP = 0.877

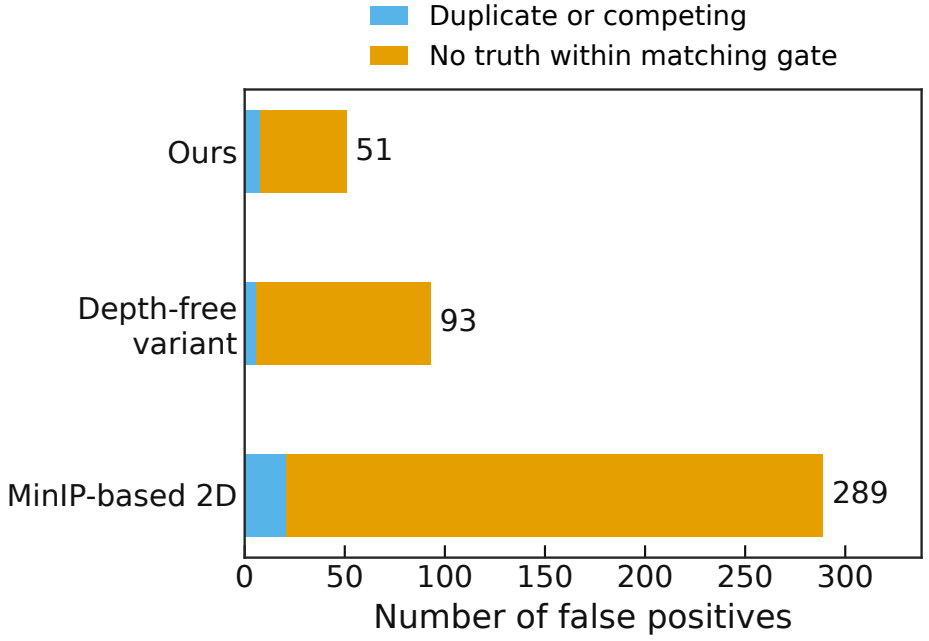
(a)


Top-N metric value
Reviewed candidates, N
Ours
Depth-free variant
MinIP-based 2D
Precision
Recall

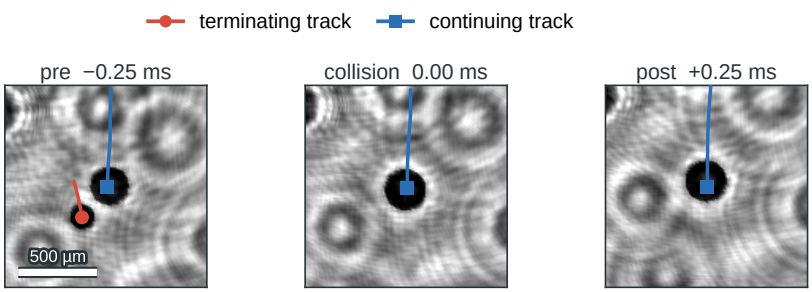
(b)


Number of false positives
Target recall
Ours
Depth-free variant
MinIP-based 2D

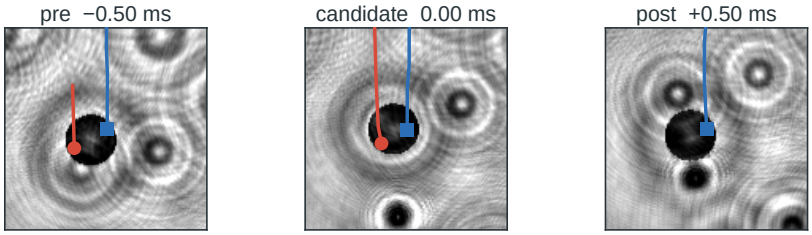
(c)


Duplicate or competing
No truth within matching gate
Ours
Depth-free variant
MinIP-based 2D
51
93
289
Number of false positives

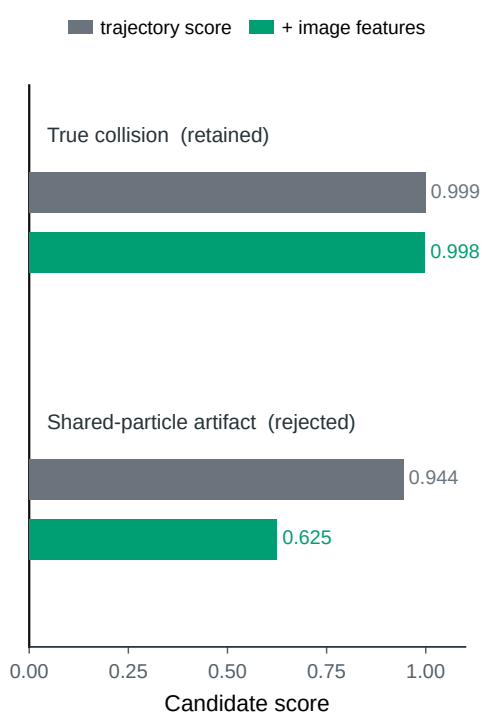
(d)



(a) Ground-truth collision: 154 + 242 → 261 µm

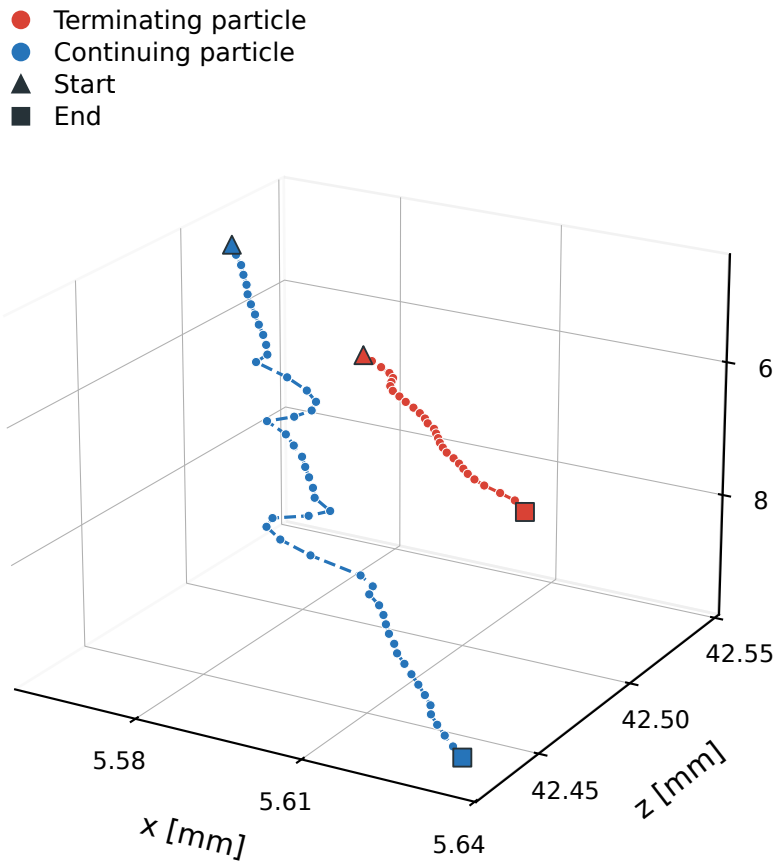


(b) Shared-particle artifact: two track IDs share one true particle

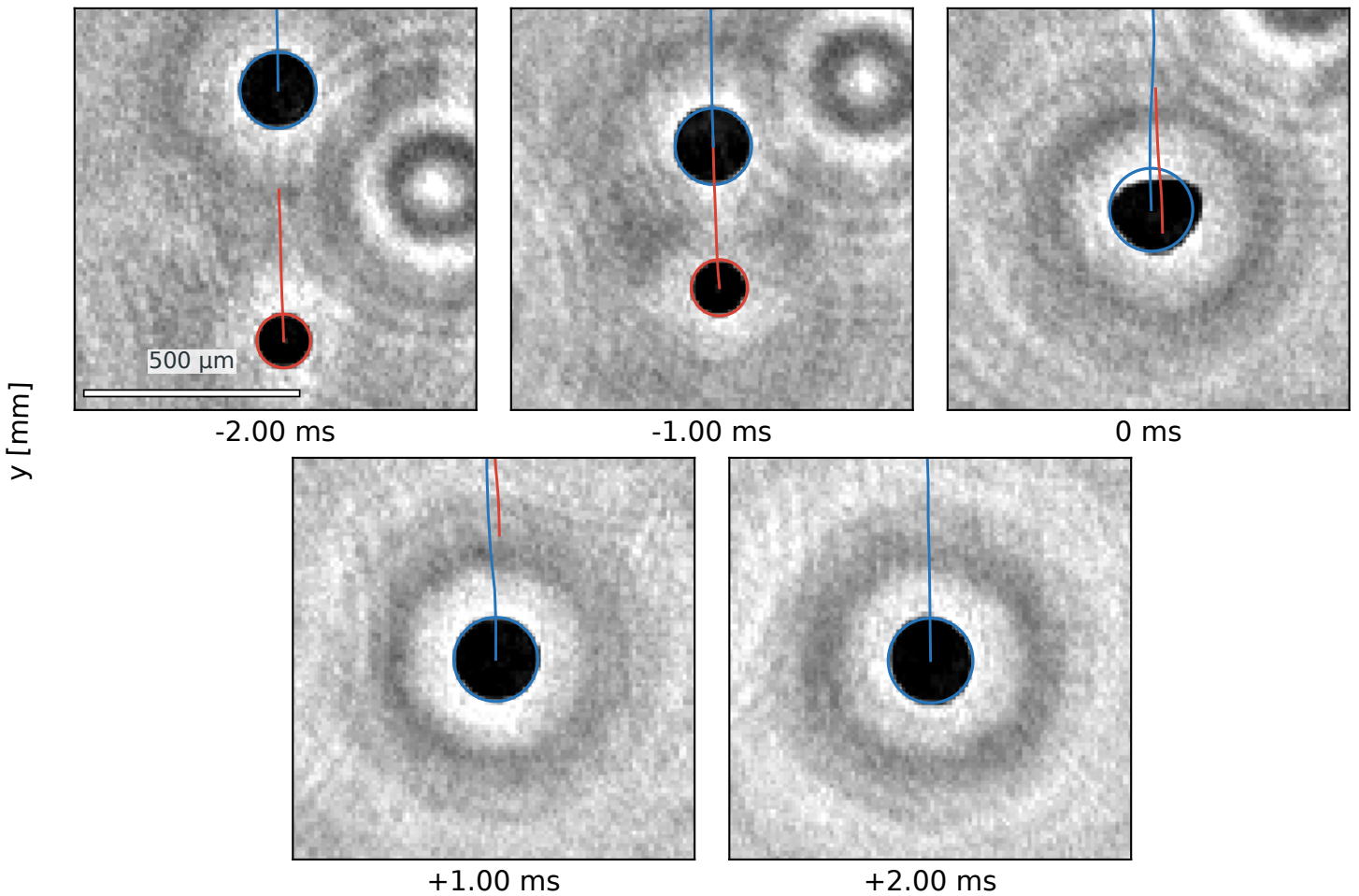


(c) Scores with and without image features

127.9 + 182.4 → 201.2 µm

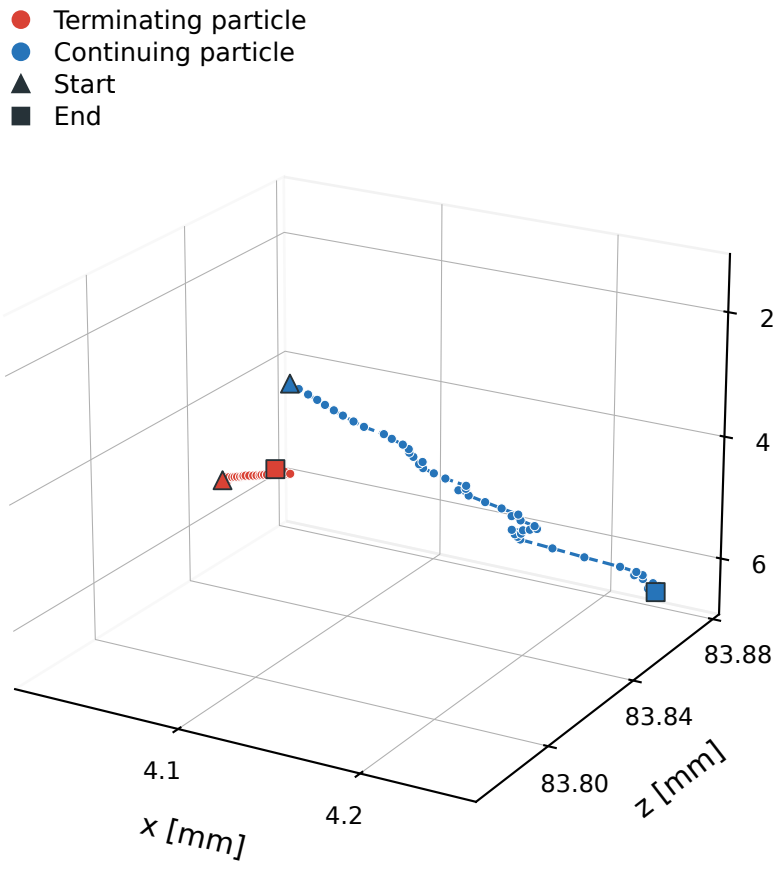


Smoothed three-dimensional trajectories

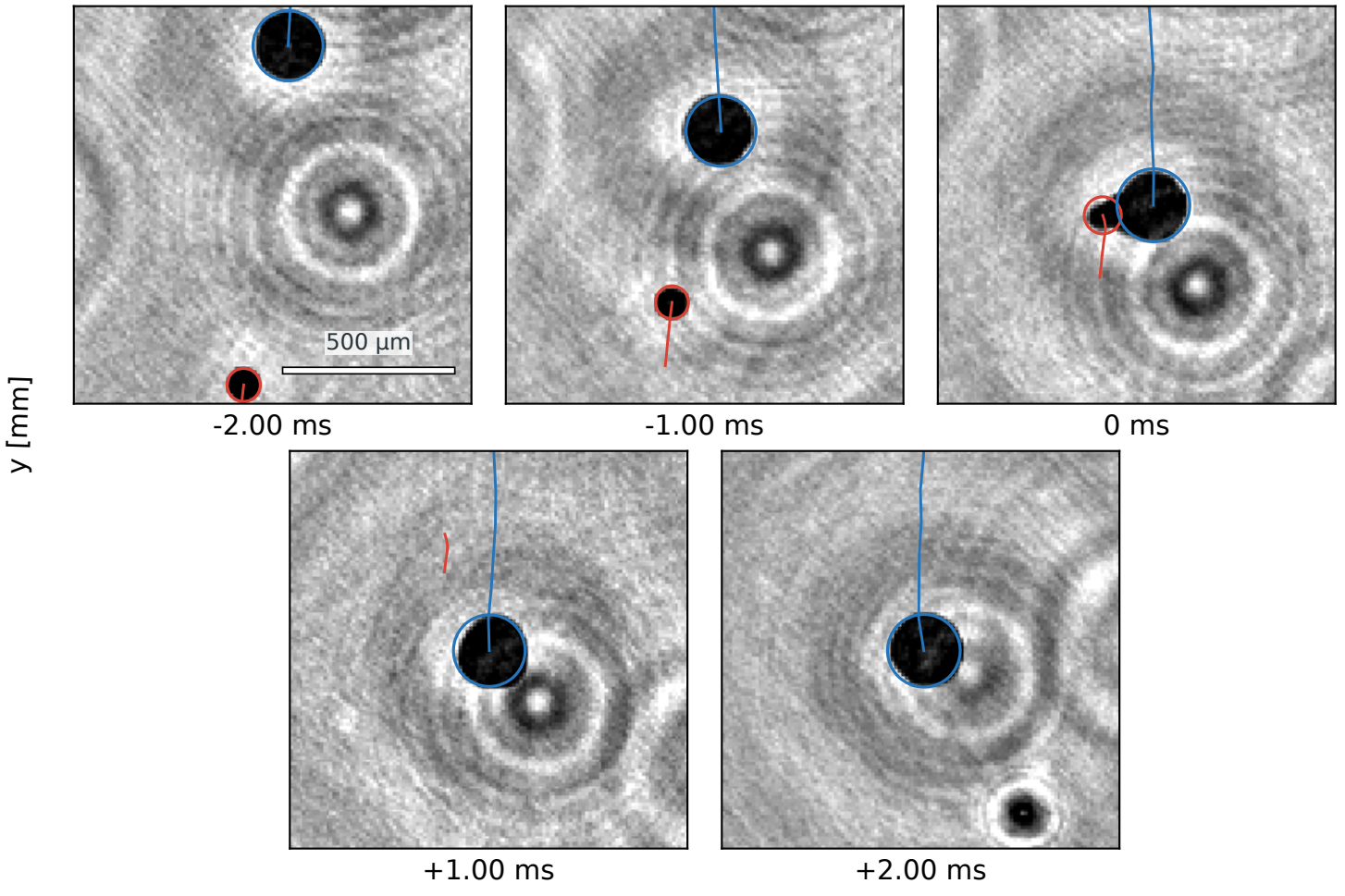


(a)

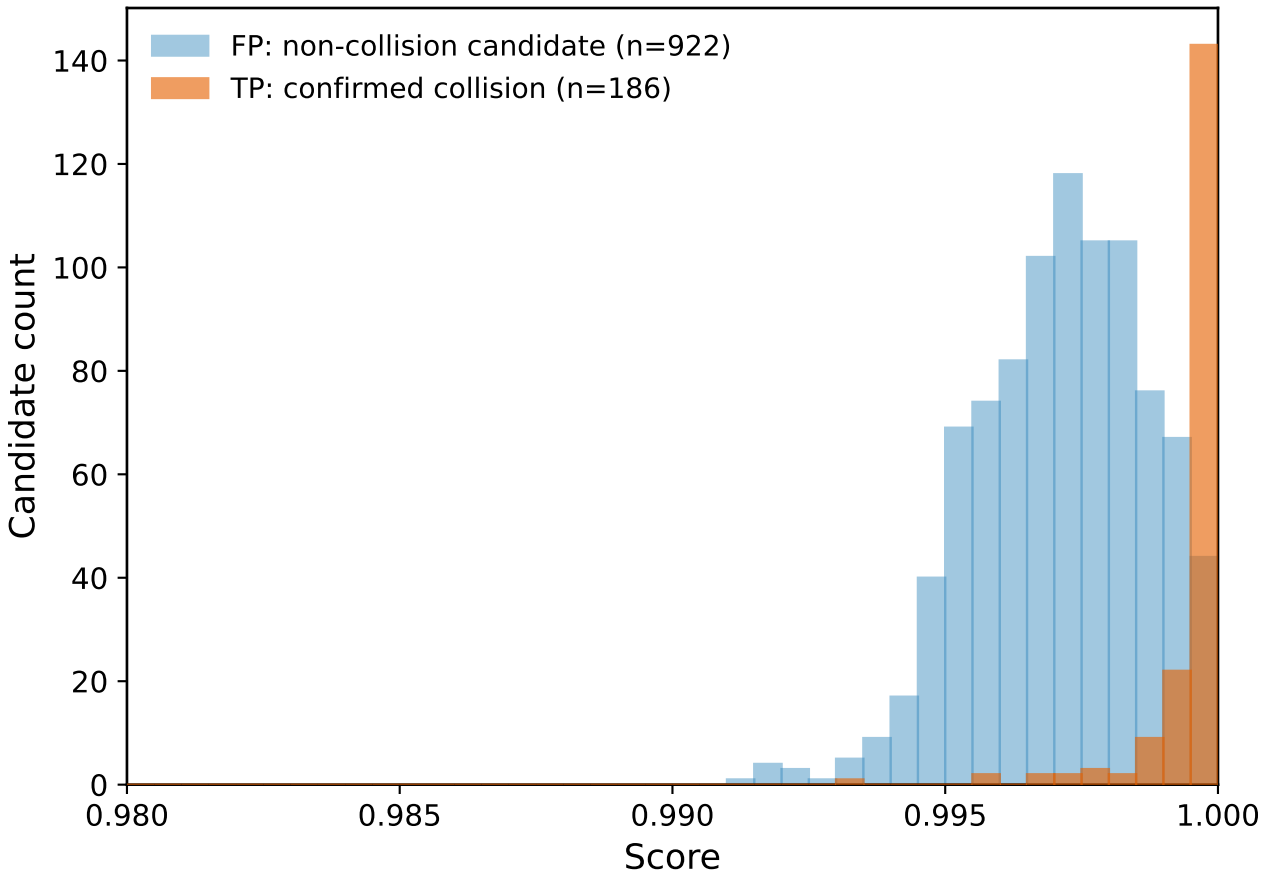
99.5 + 211.3 → 217.5 µm

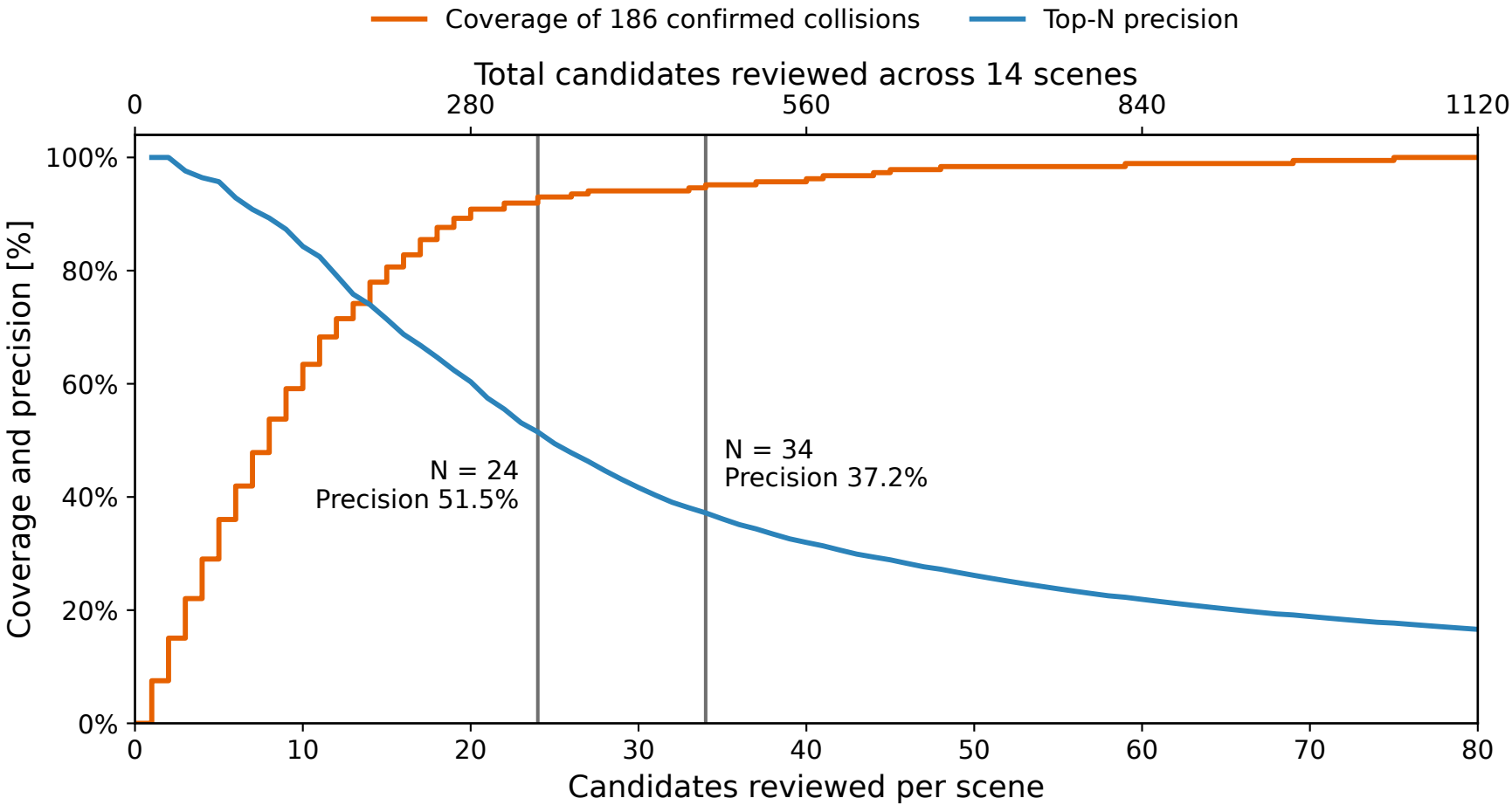


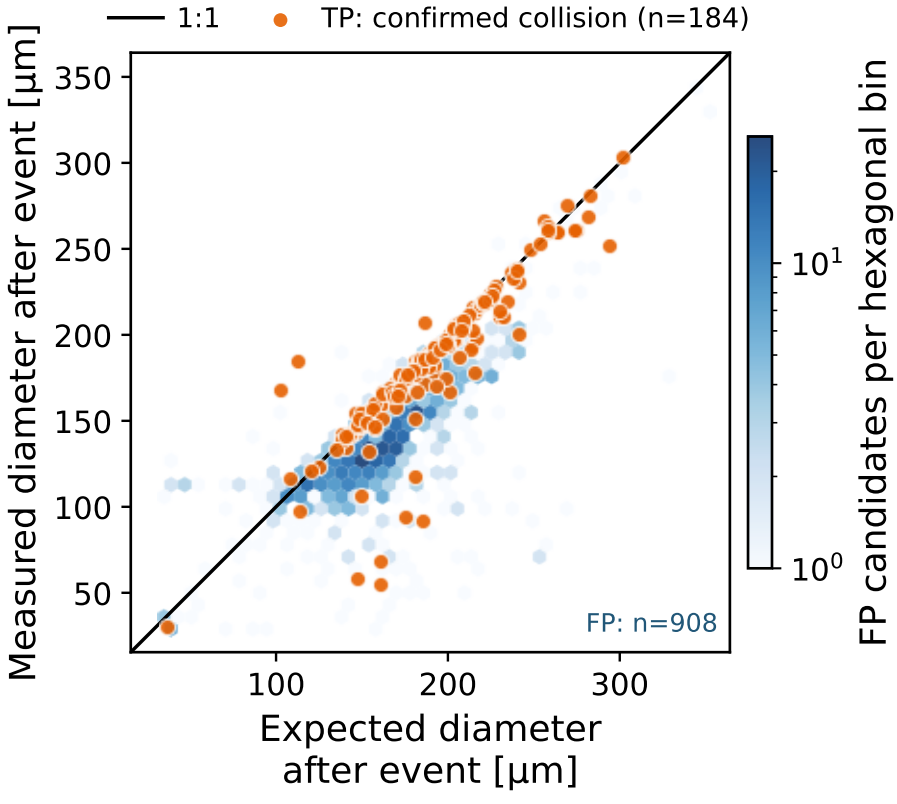
Smoothed three-dimensional trajectories



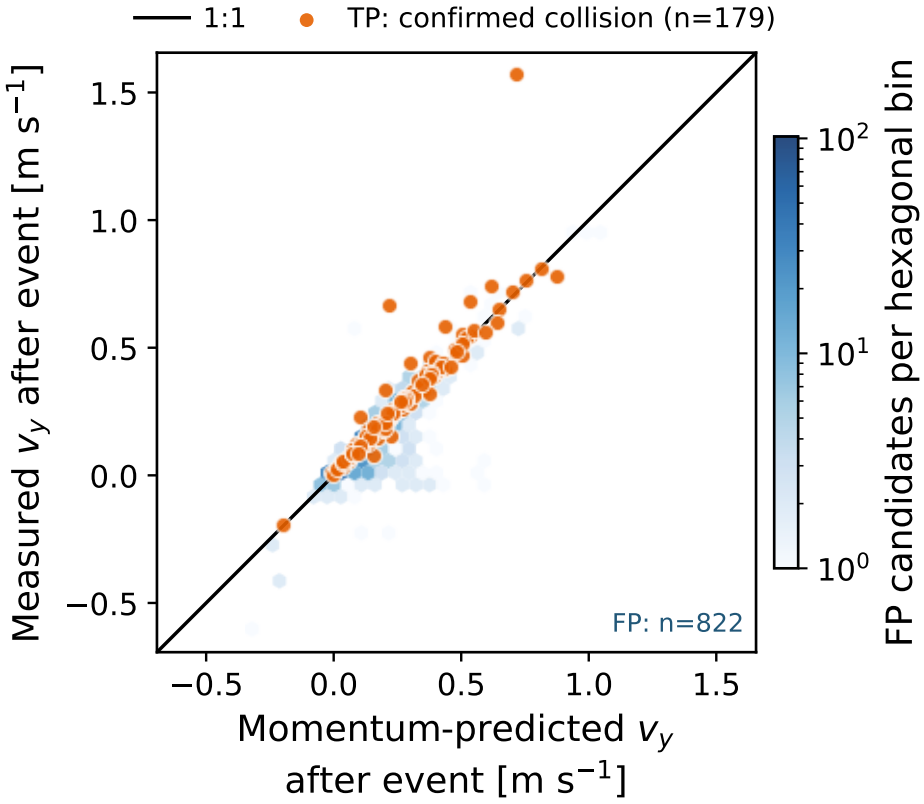
(b)


FP: non-collision candidate (n=922)
TP: confirmed collision (n=186)
Candidate count
0
20
40
60
80
100
120
140
0.980
0.985
0.990
0.995
1.000
Score

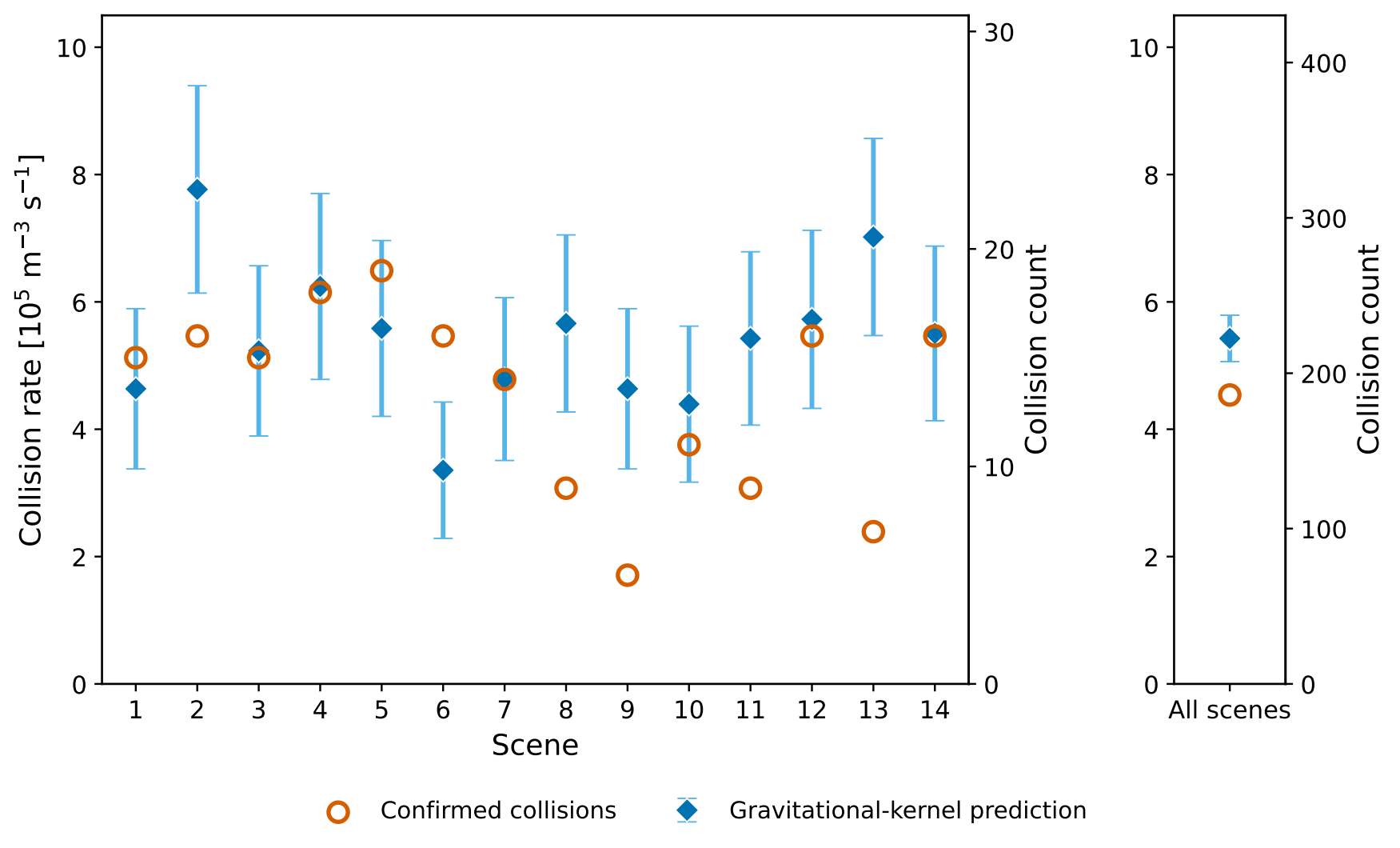

Coverage of 186 confirmed collisions
Top-N precision
Total candidates reviewed across 14 scenes
0
280
560
840
1120
100%
80%
60%
40%
20%
0%
Coverage and precision [%]
N = 24
Precision 51.5%
N = 34
Precision 37.2%
0
10
20
30
40
50
60
70
80
Candidates reviewed per scene

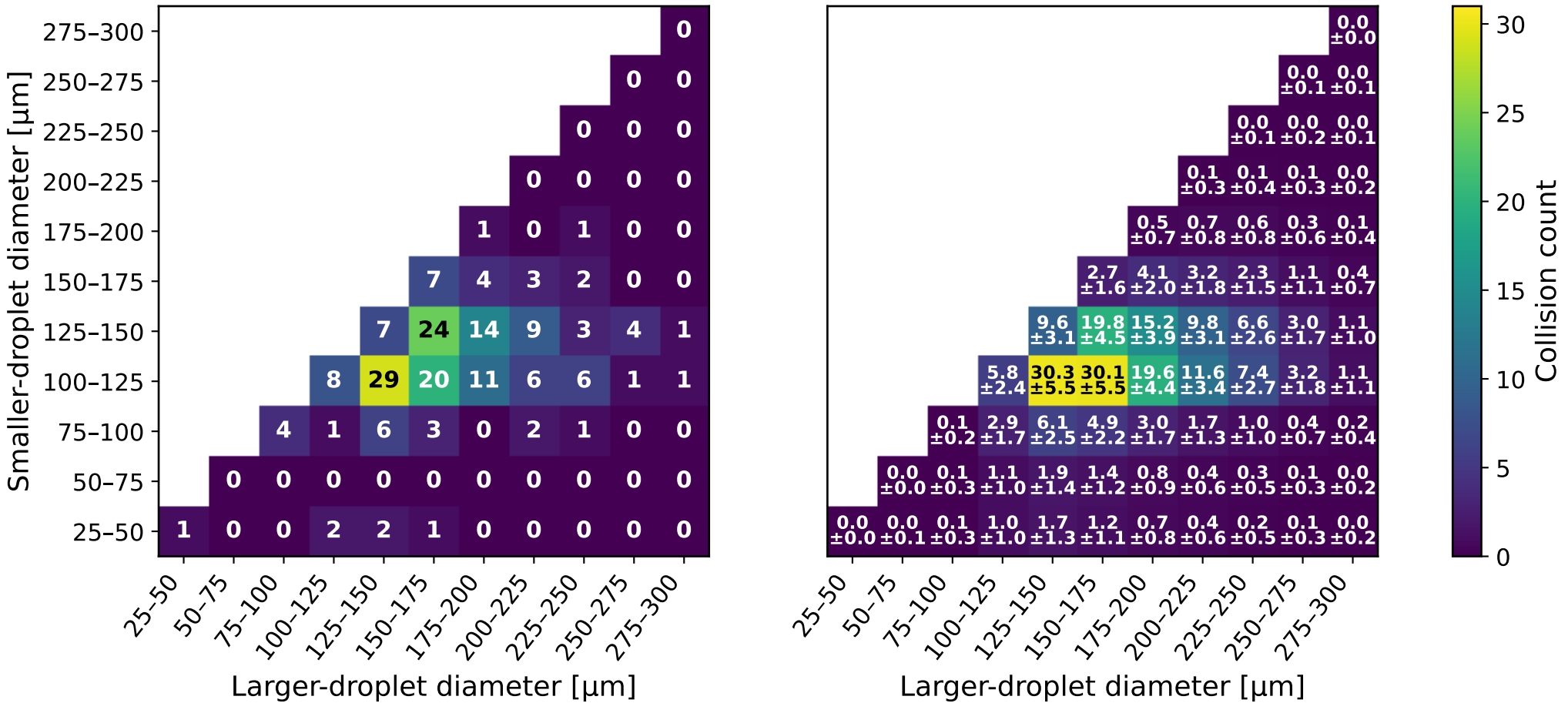


(a) Post-event diameter consistency



(b) Post-event vertical-velocity consistency


Collision rate [10^5 m^{-3} s^{-1}]
Collision count
Scene
1
2
3
4
5
6
7
8
9
10
11
12
13
14
All scenes
Confirmed collisions
Gravitational-kernel prediction


Smaller-droplet diameter [µm]
275–300
250–275
225–250
200–225
175–200
150–175
125–150
100–125
75–100
50–75
25–50
0
0 0
0 0 0
0 0 0 0
1 0 1 0 0
7 4 3 2 0 0
7 24 14 9 3 4 1
8 29 20 11 6 6 1 1
4 1 6 3 0 2 1 0 0
0 0 0 0 0 0 0 0 0 0
1 0 0 2 2 1 0 0 0 0 0
25–50
50–75
75–100
100–125
125–150
150–175
175–200
200–225
225–250
250–275
275–300
Larger-droplet diameter [µm]
(a) Confirmed collisions
0.0 ±0.0
0.0 ±0.1 0.0 ±0.1
0.0 ±0.1 0.0 ±0.2 0.0 ±0.1
0.1 ±0.3 0.1 ±0.4 0.1 ±0.3 0.0 ±0.2
0.5 ±0.7 0.7 ±0.8 0.6 ±0.8 0.3 ±0.6 0.1 ±0.4
2.7 ±1.6 4.1 ±2.0 3.2 ±1.8 2.3 ±1.5 1.1 ±1.1 0.4 ±0.7
9.6 ±3.1 19.8 ±4.5 15.2 ±3.9 9.8 ±3.1 6.6 ±2.6 3.0 ±1.7 1.1 ±1.0
5.8 ±2.4 30.3 ±5.5 30.1 ±5.5 19.6 ±4.4 11.6 ±3.4 7.4 ±2.7 3.2 ±1.8 1.1 ±1.1
0.1 ±0.2 2.9 ±1.7 6.1 ±2.5 4.9 ±2.2 3.0 ±1.7 1.7 ±1.3 1.0 ±1.0 0.4 ±0.7 0.2 ±0.4
0.0 ±0.0 0.1 ±0.3 1.1 ±1.0 1.9 ±1.4 1.4 ±1.2 0.8 ±0.9 0.4 ±0.6 0.3 ±0.5 0.1 ±0.3 0.0 ±0.2
0.0 ±0.0 0.0 ±0.1 0.1 ±0.3 1.0 ±1.0 1.7 ±1.3 1.2 ±1.1 0.7 ±0.8 0.4 ±0.6 0.2 ±0.5 0.1 ±0.3 0.0 ±0.2
25–50
50–75
75–100
100–125
125–150
150–175
175–200
200–225
225–250
250–275
275–300
Larger-droplet diameter [µm]
(b) Gravitational-kernel prediction
Collision count
30
25
20
15
10
5
0